# ASSIGNMENT 02-PROJECT 1

## Task 1:

## Team Members Selection:

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# Software Requirements Specification (SRS)

## Project Title:

## AI Meeting Summarizer & Query System

## 1. Preface

This document presents the Software Requirements Specification (SRS) for the AI Meeting Summarizer and Query System. The purpose of this document is to describe the functional and non-functional requirements of the proposed system before development begins.

The system utilizes machine learning models to transcribe and translate meeting audio conducted in English, Roman Urdu, or a mix of both languages. Based on the input, the system generates a structured English summary of the meeting and allows users to ask natural language questions about the meeting content.

This document defines the system requirements, user requirements, and system models to ensure that the project is clearly understood by all stakeholders before implementation.

## 2. Introduction

### 2.1 Purpose

The purpose of this project is to develop an application that uses machine learning models to assist teams and organizations in capturing, understanding, and querying information from meetings.

Users will be able to upload a recorded meeting audio file in English, Roman Urdu, or a combination of both. The system will transcribe the audio, translate all non-English content into English, generate a structured summary of the meeting, and allow users to ask questions about what was discussed.

The system aims to help managers, students, and research teams avoid losing important decisions and action items from meetings, without requiring any manual note-taking.

### 2.2 Scope

The AI Meeting Summarizer and Query System will allow users to process recorded meeting audio and extract structured information from it.
The system will accept the following input:

- Audio files of meetings recorded in English, Roman Urdu, or a mix of both languages within the same conversation

The system will process audio using a speech recognition model capable of understanding both English and Urdu. A language model will then be used for translation, summarization, and query answering.
All output produced by the system — including transcripts, summaries, and query responses — will be delivered entirely in English regardless of the language spoken in the meeting.
The system will support audio files in MP3, WAV, M4A, and OGG formats. Requests for other file formats will not be supported.
The full processing time is expected to take 1 to 15 minutes depending on the length of the meeting audio.
The goal of the system is to provide an accurate and queryable record of any meeting, rather than a real-time live transcription service.

# 3. Glossary

| Term | Description |
|---|---|
| **AI Model** | A machine learning model used to process inputs and generate outputs |
| **Speech Recognition Model** | A model that converts spoken audio into written text and supports multiple languages |
| **Language Model** | An AI model used for translation, summarization, and query answering |
| **Transcription** | Converting spoken audio into written text |
| **Translation** | Converting Urdu or Roman Urdu text into English |
| **Summarization** | Condensing a long transcript into key points and decisions |
| **Roman Urdu** | Urdu language written using English/Latin alphabet characters |
| **Query** | A natural language question asked by the user about the meeting |
| **Chunking** | Splitting long audio files into smaller pieces for processing |
| **Action Item** | A task assigned to a person during the meeting |
| **RAG** | Retrieval Augmented Generation — technique where AI answers questions using the transcript as context |
| **Mixed Language** | A meeting where speakers alternate between English and Urdu in the same conversation |

# 4. User Requirements Definition

## 4.1 Functional Requirements

The system shall provide the following functionalities:

1. The system shall allow users to upload audio files containing recorded meeting content.
2. The system shall accept audio files in MP3, WAV, M4A, and OGG formats.
3. The system shall transcribe all speech in the uploaded audio file to text.
4. The system shall process meetings conducted in English, Roman Urdu, or a mix of both languages without requiring manual language selection from the user.
5. The system shall translate all Roman Urdu and Urdu content in the transcript into English automatically.
6. The system shall generate a structured meeting summary in English containing the meeting overview, key decisions, action items, and topics discussed.
7. The system shall allow users to type natural language questions about the meeting and receive answers in English.
8. The system shall support meeting audio files of any length including 30-minute, 1-hour, and 3-hour meetings.
9. The system shall display the full English transcript to the user after processing.
10. The system shall allow users to download the full English transcript and meeting summary as a text file.

## 4.2 Non-Functional Requirements

**Performance**: The system should process a 30-minute meeting and generate a summary within 3 minutes. A 3-hour meeting should be processed within 15 minutes or less .

**Usability**: The system should provide a simple and user-friendly interface that allows users to upload audio files and ask questions without any technical knowledge or training.

**Reliability:** The system should handle invalid or unsupported file formats without crashing and should display appropriate error messages to the user.

**Maintainability:** The system should be designed with modular components so that machine learning models and system modules can be updated independently.

**Scalability:** The system should handle audio files of any size by automatically splitting them into chunks and processing each chunk independently.

# 5. System Requirements Definition

## 5.1 Functional Requirements

1. The system shall accept audio files in MP3, WAV, M4A, and OGG formats as input.
2. The system shall automatically split audio files larger than 25MB into chunks of 10 minutes each before sending them for transcription.
3. The system shall send each audio chunk to a multilingual speech recognition model capable of understanding both English and Urdu speech.

4. The system shall merge all chunk transcriptions in correct order into a single complete transcript.
5. The system shall send the complete transcript to a language model with instructions to detect and translate all Roman Urdu and Urdu segments into English.
6. The system shall produce a final transcript that is entirely in English regardless of the language spoken in the meeting.
7. The system shall send the English transcript to a language model to generate a structured summary with four sections: Meeting Overview, Key Decisions, Action Items, and Topics Discussed.
8. The system shall store the English transcript in session memory to allow multiple questions without re-processing the audio.
9. The system shall accept a natural language question from the user, combine it with the transcript as context, and send it to a language model to generate an answer.
10. The system shall only answer questions based on content present in the meeting transcript.
11. The system shall handle model errors and retry failed requests up to 3 times before displaying an error message to the user.
12. The system shall generate a downloadable .txt file containing the full English transcript and summary when requested by the user.

## 5.2 Non-Functional Requirements

**Accuracy:** The transcription and translation should be accurate enough that the user can trust the summary and answers without re-listening to the meeting audio.

**Response Time:** The system should produce a summary within 3 minutes for a 30-minute meeting and within 15 minutes for a 3-hour meeting.

**Security** Uploaded audio files should only be used for temporary processing and should not be stored permanently on any server.

**Compatibility** The system should be accessible via modern web browsers without requiring any software installation on the user's device.

**Modularity** The system should separate the following modules:

- Audio Input and Chunking Module
- Speech Recognition and Transcription Module
- Translation Module
- Summarization Module
- Query Answering Module

This modular architecture will allow easier system maintenance and future model improvements.

# 6. System Models

## Use Case 1: Upload and Transcribe Meeting Audio

**Actor:** User
**Description:** The user uploads a recorded meeting audio file to generate a full English transcript.
**Preconditions:** The user has access to the system interface and has a recorded audio file.
**Main Flow:**

1. The user uploads an audio file (MP3, WAV, M4A, or OGG).
2. The system validates the file format.
3. The system checks the file size and splits it into chunks if necessary.
4. The speech recognition model transcribes each chunk.
5. The system merges all chunks into a single transcript.
6. The translation module converts all Urdu and Roman Urdu content to English.
7. The system displays the full English transcript to the user.

**Postconditions:** A complete English transcript of the meeting is generated and stored in session memory.1
**Alternative Flow:**

- Step 2: If the file format is not supported, the system displays an error message and prompts the user to upload a valid file.
- Step 4: If transcription fails, the system retries up to 3 times before displaying an error.

## Use Case 2: Generate Meeting Summary

**Actor:** User
**Description:** The user requests a structured summary of the meeting after the transcript has been generated.
**Preconditions:** The meeting transcript has been successfully generated and stored in session memory.
**Main Flow:**

1. The user clicks the Generate Summary button.
2. The system retrieves the stored English transcript from session memory.
3. The language model generates a structured summary.
4. The system displays the summary with four sections: Meeting Overview, Key Decisions, Action Items, and Topics Discussed.

**Postconditions:** A structured meeting summary in English is displayed to the user.
**Alternative Flow:**

- Step 3: If the transcript is too long, the system splits it into sections, summarizes each section, and combines them into a final summary.

## Use Case 3: Ask Question About Meeting

**Actor:** User

**Description:** The user asks a natural language question about the meeting content and receives an answer based on the transcript.

**Preconditions:** The meeting transcript is stored in session memory.

**Main Flow:**

1. The user types a question in the query input box.
2. The user clicks the Ask button.
3. The system combines the question with the transcript as context.
4. The language model generates an answer based only on the meeting content.
5. The system displays the answer to the user.
6. The user may ask additional questions without re-processing the audio.

**Postconditions:** The user receives a direct and accurate answer in English from the meeting content.

**Alternative Flow:**

- Step 4: If the topic was not discussed in the meeting, the system responds "This topic was not mentioned in the meeting."

## Use Case 4: Download Transcript and Summary

**Actor:** User

**Description:** The user downloads the full English transcript and meeting summary as a text file.

**Preconditions:** The transcript and summary have been successfully generated.

**Main Flow:**

1. The user clicks the Download button.
2. The system generates a .txt file containing the full transcript and summary.
3. The file is downloaded to the user's device.

**Postconditions:** The user has a permanent local copy of the meeting transcript and summary.

# 7. Requirements Elicitation and Analysis

## 7.1 Elicitation Technique

To discover and refine the requirements for this project, the following techniques were used:
**Brainstorming** .Brainstorming sessions were conducted within the development team to explore how machine learning could assist in solving real communication problems in Pakistani workplaces and universities. Through these discussions, the idea of automatically transcribing, translating, and summarizing meetings conducted in mixed English and Urdu was identified as a practical and impactful application.

**Analysis of Existing Systems** Existing tools that assist with meeting transcription and summarization were studied to understand their functionality and limitations. Tools such as Otter.ai, Fireflies.ai, and Microsoft Teams transcription were analyzed. These tools perform adequately for English-only meetings but fail entirely when speakers switch between English and Roman Urdu, which is the dominant communication style in Pakistani organizations and universities. This gap confirmed the need for our system and directly informed the multilingual requirements.

These techniques were appropriate because they allowed the team to identify a real problem from personal experience and validate it by analyzing the shortcomings of existing solutions.

## 7.2 Stakeholder Identification

## University Students and Project Teams

Pain points:

- Group meetings for projects are rarely documented properly
- Decisions made in one session are forgotten by the next meeting
- Members who miss a meeting have no way to catch up except asking others
- Meetings naturally mix Urdu and English which no existing tool supports

The system helps students by automatically capturing all decisions and allowing anyone to ask what was discussed.

## Corporate Managers and Teams

Pain points:

- Action items from meetings are often not followed up because they were never properly recorded
- Managers spend time re-explaining decisions from previous meetings
- Meeting minutes written manually are slow, incomplete, and inconsistent

The system helps managers by automatically generating action items and decisions immediately after every meeting.

## Research Teams

Pain points:

- Research discussions contain important ideas that need to be retrieved later
- Long brainstorming sessions produce valuable content that gets lost over time
- No easy way to search what was discussed in a meeting from weeks ago

The system helps research teams by enabling them to ask specific questions about any past meeting instantly.

## How this system addresses their pain points:

- Automatic transcription removes the need for a dedicated note-taker

- Full support for mixed English and Urdu speech means users do not need to change how they naturally communicate
- All output is delivered in English regardless of input language
- The query system allows anyone to ask what was decided instead of re-listening to hours of audio
- Action item extraction ensures nothing assigned in the meeting is forgotten
- Downloadable output means the record is preserved permanently

# Task 4: Streamlit Demo:

## Link of Colab Notebook

**https://colab.research.google.com/drive/1B7Sy1c87VUfrtX-J7-06H0N31pu5fuey?usp=sharing**

## URL of Streamlit Demo:

**https://suzi-prerighteous-bellicosely.ngrok-free.dev/**

**Grok api key**: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

# Task 5:

### Handwritten Document:

Date: _____ Day: _____

# Assignment 02

## Project 1

* Project Introduction:-

The purpose of this project is to develop an application that uses machine learning models to assist teams and organization in capturing, understanding and querying informations from meetings. Users will be able to upload a recorded meeting audio file in English, Roman Urdu or combination of both. The system will transcribe audio, translate both English content into English, generate a structured summary of the meeting and allow users to ask questions about what was discussed.

* Use Case: Generate Meeting Summary
* Actor: User
* Description: The user request a structured summary of the meeting after transcript has been generated.
* Preconditions:

The meeting transcript has been succesfully generated and stored in session memory.

* Flow:

• Users click the generate summary button. The system retrieves transcript. System employs summary with 4 sections: Key decisions, Action items and topics discussed.

* Post Conditions: A structured meeting summary in English and is displayed to the user.